

AGENT RUNTIME COST & RISK AUDIT REPORT

Generated: 2026-06-22 11:35:08 UTC | Behavioral Efficiency Score: 88/100
(Deterministic Baseline)

🔗 **Metric Calculation Formula:** Score = \$100 - \sum(ext{Deductions})\$. Impact
Matrix: RetryLoop = -25 | RepeatedRead = -15 | RapidScan = -8 |
ProtectedConfigMutation = -4.

Section 1 — Executive Summary

Audited session cost: \$0.831
Audited waste footprint: \$0.069 (8% of session cost)
Monthly projection: \$103.72 wasted at 50 sessions/day
Critical alerts: 1 protected target modification alert
Action required: 2 items (see Technical Findings below)

Section 2 — Technical Findings

2.1 Budget & Efficiency Telemetry

Evaluates agent processing behavior against runaway API consumption and token waste patterns.

- Runtime Health Status: ● OPTIMIZED

Audited Financial Metric	Value	Context
Session Cost (audited)	\$0.8308	Real-time computed API credit draw.
Audited Volumetric Waste	\$0.0691	8% of session runtime consumed by infractions.
Projected Monthly Waste	\$103.72	Scaled footprint at 50 developer sessions/day.

2.2 Behavioral Cost Indicators

Observed Waste Pattern	Target Count	Financial & Operational Implication
Estimated Wasteful Operations	4	Volumetric measure of redundant processing activity.
Retry Loop Incidents	0	Runaway compute waste; agent is stuck executing broken pipelines.
Repeated Target Reads	0	Context window waste; agent

Observed Waste Pattern	Target Count	Financial & Operational Implication
		mismanaged previously read details.
Rapid Enumeration Bursts	1	Elevated context acquisition overhead from excessive repository scanning.
Protected Config Mutations	1	Critical corporate infrastructure configuration mutation risks.

2.3 Resource Allocation Ledger

- 👉 **Context Gathering (READ):** 8 file ingestion operations.
- State Modifications (WRITE): 6 codebase mutation footprints.
- ⚙️ **Runtime Execution (BASH):** 4 environment pipeline requests.

2.4 Itemized Cost & Risk Breakdowns

[1] Excessive Repository Context Scan - RESOURCE_DRAIN (MEDIUM)

- Impact Asset Target:** Multiple Files (main.py, models.py, routers/auth.py)
- Triggering Samples:** 3 wasteful iterations inside time window constraint
- Tracked Window Cost Incurred:** 10639 tokens burned
- Defensible Forensic Evidence:**
 - 2025-06-20T11:14:36.200Z
 - 2025-06-20T11:14:39.300Z
 - 2025-06-20T11:14:41.800Z

[2] Protected Configuration Modification - COMPLIANCE_VIOLATION (HIGH)

- Impact Asset Target:** .env
 - Triggering Samples:** 1 wasteful iterations inside time window constraint
 - Tracked Window Cost Incurred:** 12410 tokens burned
 - Defensible Forensic Evidence:**
 - [Target Match] Agent created/modified a credential file: .env
-

2.5 Recommended Action Items

Recommendation #1

🔗 **Source Provenance Citation:** Linked to engine anomaly footprint detected by `RapidFileEnumerationDetector` on resource target `Multiple Files` (`main.py`, `models.py`, `routers/auth.py`).

1. **Observed Behavior:** The system detected an 'Excessive Repository Context Scan' through the 'RapidFileEnumerationDetector'. This behavior was observed across 3 incidents, specifically impacting 'main.py', 'models.py', and 'routers/auth.py' at timestamps 2025-06-20T11:14:36.200Z, 2025-06-20T11:14:39.300Z, and 2025-06-20T11:14:41.800Z. This indicates the LLM application is accessing or enumerating a broader set of files than typically required for its immediate task.
2. **Potential Impact:** Unnecessary file enumeration can lead to increased I/O operations, higher computational resource consumption, and extended processing times, directly impacting application latency and operational costs. Furthermore, it increases the risk of inadvertently exposing sensitive or irrelevant information to the LLM's context, potentially leading to compliance violations or model performance degradation due to irrelevant data.
3. **Suggested Review Action:** Evaluate the application's context assembly mechanisms to ensure that only explicitly required files and directories are included in the LLM's operational scope. Implement precise directory exclusion rules or define task-specific scopes to limit file access.

Suggested Strategic Review:

Implement strict context management policies by defining explicit directory exclusion rules and establishing task-specific scopes for file access. This involves configuring the application to only provide the LLM with access to the minimal set of files necessary for its current task, thereby preventing broad repository scans.

Recommendation #2

🔗 **Source Provenance Citation:** Linked to engine anomaly footprint detected by `SecretAccessDetector` on resource target `.env`.

1. **Observed Behavior:** An agent was detected creating or modifying a credential file named '.env'.2. **Potential Impact:** Unauthorized modification of protected configuration files can lead to security vulnerabilities, data exposure, or system instability. This directly impacts the integrity of sensitive operational parameters.3. **Suggested Review Action:** Review the agent's assigned operational scope and file system access permissions to ensure it cannot interact with sensitive configuration files.

Suggested Strategic Review:

Implement strict directory exclusion rules or refine the agent's task-specific scope to prevent access to and modification of sensitive configuration files and directories.